METRIC
MIL-F-53080
28 January 1988

MILITARY SPECIFICATION

FUEL, ENGINE DESIGN, REFEREE GRADE,

TYPE I AND TYPE II

(METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This multifuel engine design referee grade fuel specification covers two types of referee diesel fuels (see 1.2) for use in research, development, and engineering of next generation engine powerplant systems being designed for both combat/tactical vehicles and equipment, and mobile electric power generators that will be required to operate on a considerably wider range of fuels which will not conform to VV-F-800 (see 6.1). Engines that demonstrate satisfactory operability on both fuel types will be capable of operating on a wide range of commercially available fuels as well as those fuels of lower quality anticipated to be marketed in the 1990-2025 time frame.

1.2 Classification.

Type I - High volatility referee diesel fuel.
Type II - Low volatility referee diesel fuel.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9140

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2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-I-25017	- Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble.
MIL-S-53021	- Stabilizer Additive, Diesel Fuel.

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets, Preparation and the Submission of.
FED-STD-791	- Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-290	- Packaging, Packing, and Marking of Petroleum and Related Products.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DOL)

OSHA 29CFR1910.1200 Hazard Communication Interpretation Regarding Lubricating Oils.

(Guideline CPL 2-2.38 may be obtained from OSHA Publications Office, Room S-4203, 200 Constitution Avenue, NW, Washington, DC 20210.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue

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of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D 86 - Distillation of Petroleum Products.
D 93 - Flash Point by Pensky-Martens Closed Tester.
D 129 - Sulfur in Petroleum Products (General Bomb Method).
D 130 - Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test (Note: Use the ASTM Copper Strip Corrosion Standard for Corrosion).
D 323 - Vapor Pressure of Petroleum Products (Reid Method).
D 445 - Kinematic Viscosity of Transparent or Opaque Liquids (and the Calculation of Dynamic Viscosity).
D 482 - Ash from Petroleum Products.
D 524 - Ramsbottom Carbon Residue of Petroleum Products.
D 613 - Ignition Quality of Diesel Fuels by the Cetane Method.
D 1298 - Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method).
D 1552 - Sulfur in Petroleum Products (High-Temperature Method).
D 2274 - Oxidation Stability of Distillate Fuel Oil (Accelerated Method).
D 2276 - Particulate Contaminant in Aviation Turbine Fuels.
D 2386 - Freezing Point of Aviation Fuels.
D 2500 - Cloud Point of Petroleum Oils.
D 2551 - Vapor Pressure of Petroleum Products (Micromethod), Test for.
D 2622 - Sulfur in Petroleum Products (X-Ray Spectrographic Method).
D 3241 - Thermal Oxidation Stability of Aviation Turbine Fuels.
D 3343 - Hydrogen Content of Aviation Fuels, Estimation of.
D 3701 - Hydrogen Content of Aviation Turbine Fuels by Low Resolution Nuclear Magnetic Resonance Spectrometry.
D 4057 - Manual Sampling of Petroleum and Petroleum Products.
D 4176 - Free Water and Particulate Contamination of Distillate Fuels (Clear and Bright Pass/Fail Proceedings).
D 4177 - Automatic Sampling of Petroleum and Petroleum Products.

(The test methods listed above are included in Volumes 05.01, 05.02, 05.03, and 05.04 of the Annual Book of ASTM Standards and are available individually. Applications for copies of all ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this

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specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The diesel fuels shall be petroleum distillates and shall contain catalytically or thermally processed blending fractions with additives as specified in 3.2.

3.2 Additives.

3.2.1 Stabilizer additive. The finished fuel shall contain stabilizer additives in accordance with MIL-S-53021. Additives shall be used at concentrations approved during the qualification process (see 6.4).

3.2.2 Corrosion inhibitor. The finished fuel shall contain corrosion inhibitor/lubricity improver additives in accordance with MIL-I-25017. Additives shall be used at concentrations approved during the qualification process (see 6.4).

3.2.3 Cetane improver. The cetane requirements may be obtained by using any of the listed compounds. Any one or any combination of cetane improvers shall be not greater than 0.50 percent by weight.

a. Amyl nitrate.
b. Hexyl nitrate.
c. Octyl nitrate.
d. Structural isomers of a., b., and c.
e. Isopropyl nitrate.
f. Cyclohexyl nitrate.

3.3 Physical and chemical requirements. The finished diesel fuels shall be in accordance with table I.

TABLE I. Physical and chemical requirements.

Property	Type I	Type II
Distillation, °C		
Initial boiling point	NR[1]/	Report
10% V recovered at,	127 max	Report
50% V recovered at,	191 max	Report
90% V recovered at,	232 max	316 to 385
End point	288 max	Report
Density, Kg/L @15 °C	0.750 to 0.801	0.934 max
(Gravity, °API)	(45 to 57)	(20 min)
Reid vapor pressure, kPa	6.9 to 34.5	NR
(psig)	(1 to 5)	

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TABLE I. Physical and chemical requirements. (Cont'd)

Property	Type I	Type II
Flash point, °C, min	NR	54
Cetane number	20 to 30	35 max
Carbon residue on 10% bottoms, mass % max	0.20	0.40
Kinematic viscosity, cSt at 40 °C	0.9 max	7.0 to 9.0
Sulfur, wt % 2/	0.50 max	0.80 to 1.20
Ash, wt % max	0.01	0.05
Hydrogen, % wt	Report	NR
Cloud point, °C, max	NR	-7
Freezing point, °C, max	-40	NR
Thermal stability, JFTOT at 260 °C for 2.5 Hr.		
Pressure drop, millimeters Hg	40 max	NR
Tube rating	3	NR
Accelerated stability, mg/100 mL, max	NR	1.5
Particulate content, mg/L, max	10	2.0
Copper corrosion, 3 hr. 50 °C, max	No. 1	No. 1

1/ Not required.
2/ Naturally occurring sulfur is preferred, the addition of Tertiary Butyl DiSulfide (TBDS) is permitted. Naturally-occuring sulfur shall be greater than half of the total sulfur in the finished fuel.

3.4 Material safety data sheets. The material data sheets shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200. When FED-STD-313 is at variance with the CFR, 29 CFR 1910.1200 shall take precedence, modify and supplement FED-STD-313.

3.5 Workmanship. The finished fuels shall be visually free from undissolved water, sediment, and suspended matter; and shall be clear and bright at the ambient temperature or 21 °C, whichever is higher.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling.

4.2.1 Sampling for the examination of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105, at inspection level II.

4.2.2 Sampling for tests. Take samples from bulk or packaged lots for tests in accordance with ASTM D 4057 or ASTM D 4177.

4.3 Inspection. Perform inspection in accordance with method 9601 of FED-STD-791.

4.3.1 Examination of filled containers. Examine samples taken in accordance with 4.2.1 for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Reject any container having one or more defects, or under the required fill. If there are defective or underfilled containers, reject the lot represented by the sample.

4.4 Test methods. Perform tests in accordance with the methods listed in table II.

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TABLE II. Test methods.

Test	ASTM Method Number
Distillation	D 86
Density, API Gravity	D 1298
Reid vapor pressure 1/	D 323 or D 2551
Flash point	D 93
Cetane number	D 613
Carbon residue on 10% bottoms	D 524
Kinematic viscosity	D 445
Sulfur 1/ 2/	D 1552, D 129, or D 2622
Ash	D 482
Hydrogen 1/	D 3343 or D 3701
Cloud point	D 2500
Freezing point	D 2386
Thermal stability, JFTOT	D 3241
Accelerated stability	D 2274
Particulate content	D 2276
Copper corrosiveness	D 130
Workmanship 3/	D 4176

1/ Report which ASTM test method was used for the determination.
2/ D 1552 is the preferred method for determining sulfur, but either D 129 or D 2622 may be used as an alternate.
3/ Method A or B shall be used.

4.5 Report of analysis. The contractor shall make available copies of analysis reports, giving the table I test results.

4.6 Additives used. The contractor shall make available copies of the documents explaining the types and concentrations of the additives blended to the diesel fuel.

4.7 Inspection of packing. Packing of the diesel fuel shall be examined for conformance with the applicable requirements of MIL-STD-290.

5. PACKING

5.1 Packing and marking. Pack and mark the diesel fuel in accordance with MIL-STD-290 (see 6.2). Packing shall be level B, C, or commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. Type I and type II fuels are intended to provide engine designers and developers with two fuels exhibiting each end of the fuel spectrum. Engines that demonstrate satisfactory operability on both fuels will be capable of operating on a wide range of commercially available fuels as well as those

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fuels of lower quality anticipated to be marketed in the 1900-2025 time frame. These two types of fuels are not to be confused with referee diesel fuel procured under MIL-F-46162 that is used in conjunction with current production engine systems undergoing Development Testing (DTI through DTIII). MIL-F-46162 fuel is intended for use in proof-testing of compression-ignition engines, diesel-powered auxiliary units, gas turbine engine-driven ground vehicles and mobile electric power generators, and other fuel handling supply items.

6.1.1 Background information. Interim Report BFLRF No. 172 provides information on the development of the test requirements presented in this specification. Included in the report is a position statement governing the use of MIL-F-46162 referee grade diesel fuel.

U.S. ARMY BELVOIR FUELS AND LUBRICANTS RESEARCH FACILITY (SwRI)
INTERIM REPORT - Development of Referee Fuels for Improved Army Multifuel Engine Design
BFLRF No. 172
AD A171143

(Copies of this document may be obtained from National Technology Information Service (NTIS), P.O. Box 1556, Springfield, VA 22151 or Defense Technical Information Center (DTIC), Cameron Station, Alexandria, VA 22304-6145.)

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following information:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable to this contract and exceptions thereto (see 2.1.1).
c. Type of fuel required (see 1.2).
d. Quantity of fuel required.
e. Type and size of containers required (see 5.1).
f. Degree of packing required (see 5.1).

6.3 Definitions.

6.3.1 Bulk lot. A bulk lot is an indefinite quantity of a homogeneous mixture of one fuel type offered for acceptance in a single, isolated container; or manufactured in a single plant run (not to exceed 24 hours), through the same processing equipment, with no change in the ingredient materials.

6.3.2 Packaged lot. A packaged lot is an indefinite number of 55-gallon drums or smaller unit containers of identical size and type, offered for acceptance, and filled with a homogeneous mixture of one fuel type from a single, isolated container; or filled with a homogeneous mixture of one fuel type, manufactured in a single plant run (not to exceed 24 hours), through the same processing equipment, with no change in the ingredient materials.

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6.3.3 Type I referee fuel. Type I fuels resemble wide-cut aviation turbine JP-4 type fuels, in that the fuel is more naphtenic in composition, has high volatility, and low viscosity characteristics. The cetane number is low.

6.3.4 Type II referee fuel. Type II fuel is a heavier distillate with considerably lower volatility, higher sulfur content, and higher viscosity characteristics than fuels in accordance with VV-F-800. Type II fuel has a low cetane number.

6.4 Additive concentrations. Approved concentration of corrosion inhibitor and stabilizer additives are given in QPL-25017 and QPL-53021, respectively.

6.5 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with 3.4. The pertinent government mailing addresses for submission of data are listed in appendix B of FED-STD-313.

6.6 Subject term (key word) listing.

Diesel fuel
Referee diesel fuel
Type I referee fuel
Type II referee fuel

Custodians:
Army - ME
Navy - AS
Air Force - 68

Preparing activity:
Army - ME

Project 9140-A106

Review activities:
Army - AT,CE
DLA - PS

User activity:
Army - TE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-F-53080	Fuel, Engine Design, Referee Grade, Type I and Type II (Metric)

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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